

Peyton Rivers

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EDUCATION

- University of North Carolina at Charlotte (UNCC), Charlotte, NC | Graduation: December 2028
- Bachelor of Science in Computer Science | Concentration: Data Science

PERSONAL PROJECTS

Tabular California Housing Kaggle Competition

- Placed **2nd** place out of all competitors
- Researched Domain to get a better understanding for feature engineering
- Used a LightGBM Regressor model and optimize its hyper parameters without overfitting
- Used visuals and mutual info regressor to determine which features were most valuable and connected to properly Feature engineer
- Deployed Final Model into the test data set resulting with me getting 2nd place

Link: <https://github.com/peytonrivers/Kaggle-Regression---Tabular-California-Housing-Competition/tree/main>

Charlotte AI Research Project

- Developing a personalized tutor based on their student portfolio by using our school's Canvas API retrieval.(FastAPI, MySQL, Canvas API)
- My responsibility is setting up the **Database** and to build the **api** to retrieve the data from Canvas and then to store it in the database
- Working with a collaborative team, showing leadership and problem-solving within our team

Personalized Finance Power BI Dashboard

- Developed an interactive dashboard in Power BI to analyze monthly income, spending patterns, and savings trends.
- Connected Excel and SQL data sources to support dynamic updates and drill-through interactivity.
- Built custom DAX measures for KPIs such as savings rate, expense ratios, and month-over-month growth.
- Delivered visual insights through charts, slicers, and KPIs that highlight actionable personal finance metrics.

TECHNICAL SKILLS

- **Languages:** Python, SQL
- **Libraries:** Pandas, NumPy, Scikit-Learn, PyTorch
- **Tools:** Power BI, Excel, Git, Streamlit
- **Concepts:** Machine Learning, Data Preprocessing, Data Visualization, Feature Engineering, Model Evaluation, Cross-Validation

STUDENT INVOLVEMENT

Association of Data Science

- Developing Machine Learning models
- Using Data Exploration tools like pandas, Numpy, Matplotlib, and Streamlit
- Learning Machine Learning Basics and SQL skills

Charlotte AI Research (CAIR)

- Developing a personalized tutor based on each student's portfolio by using our school's Canvas API retrieval
- Active member who participates in our weekly meetings
- Learned the basic theory on LLM embeddings